

Venn Diagrams and Contingency Tables

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Events and sample spaces ([Note 330](#)).
- **Leads to:** Joint and marginal probability ([Note 341](#)).

1. Overview

Key point: A Venn diagram draws events as overlapping circles inside the sample space; a contingency table holds the same counts in a grid. Both show at a glance which outcomes two events share.

One die: $A = \text{at least } 4$, $B = \text{even}$

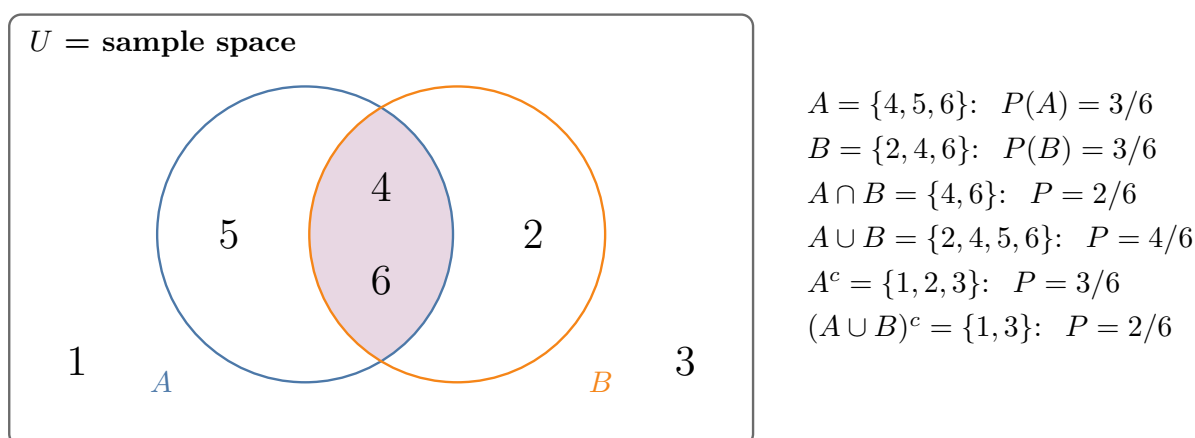


Figure 1: One die with two events, $A = \text{at least } 4$ and $B = \text{even}$, as a Venn diagram

Figure 1 rolls one die and draws two events: $A = \text{“at least 4”}$ and $B = \text{“even”}$. Every outcome sits in exactly one region: in A only, in both, in B only, or in neither. Counting the outcomes in a region gives its probability.

This Note covers two ways of showing how events overlap. The Venn diagram is a picture; the contingency table is the same information as a grid of counts. Joint, marginal and conditional probabilities, the subject of the [joint, marginal and conditional probability Note](#), are all read off these two views.

Sample spaces and events are defined in the [random experiments and events Note](#); the complement and addition rules used here are in the [empirical and theoretical probability Note](#).

2. Venn diagrams

Key point: In a Venn diagram the rectangle is the whole sample space, with probability 1; each event is a circle, and overlaps show shared outcomes.

A **Venn diagram** comes from set theory. A rectangle stands for the **universal set** U , everything there is. Each event is drawn as a circle inside it.

2.1 The rectangle is the sample space

Key point: In probability, the universal set is the sample space, so $P(U) = 1$.

In probability the universal set is the sample space of the experiment. For one die, the rectangle holds the outcomes 1 to 6.

Every trial gives some outcome inside the rectangle, so the rectangle is the sure event:

$$P(U) = 1$$

2.2 One event and its complement

Key point: A circle marks one event; everything in the rectangle outside the circle is its complement.

Take the event $A = \text{“a number of at least 4”} = \{4, 5, 6\}$. Its outcomes go inside a circle; 1, 2 and 3 stay outside it, in the rectangle.

- $P(A) = 3/6 = 1/2$: three of the six equally likely outcomes.
- The region outside the circle is the complement $A^c = \{1, 2, 3\}$. Its probability is the whole rectangle minus the circle (the complement rule):

$$P(A^c) = P(U) - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

2.3 Two events and their four regions

Key point: Two overlapping circles split the sample space into four regions: A only, both, B only and neither; each union, intersection or complement is a set of these regions.

Add a second event, $B = \text{“an even number”} = \{2, 4, 6\}$. The outcomes 4 and 6 belong to both events, so they go in the overlap of the two circles (Figure 1):

Region	Outcomes	Probability
A only	5	$1/6$
both A and B	4, 6	$2/6$
B only	2	$1/6$
neither	1, 3	$2/6$

Region	Outcomes	Probability
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The four probabilities add up to $6/6 = 1$, because every outcome lies in exactly one region. Figure 2 shades the four events built from these regions.

Four regions, four events

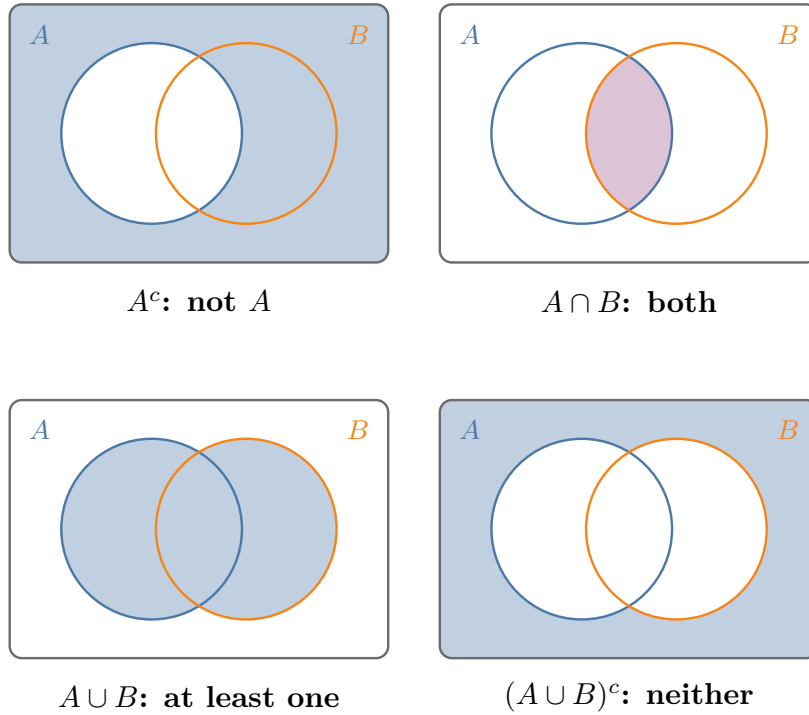


Figure 2: Four events as shaded regions of a two-circle Venn diagram

Each one is read from the diagram by counting the outcomes in its shaded part:

- **Intersection** $A \cap B$, the overlap: $\{4, 6\}$, so $P(A \cap B) = 2/6$.
- **Union** $A \cup B$, everything inside either circle: $\{2, 4, 5, 6\}$, so $P(A \cup B) = 4/6$.
- **Complement** A^c , everything outside circle A : $\{1, 2, 3\}$, so $P(A^c) = 3/6$.
- **Neither**, $(A \cup B)^c$, everything outside both circles: $\{1, 3\}$, so $P = 2/6$.

The union can also be found without counting, by the general addition rule from the [empirical and theoretical probability Note](#). The overlap is inside both circles, so it is subtracted once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6}$$

“Neither” is the complement of the union, so it follows from the union:

$$P((A \cup B)^c) = 1 - P(A \cup B) = 1 - \frac{4}{6} = \frac{2}{6}$$

Both routes agree with the counts in the table above.

Extra: “Neither A nor B ” can also be built as “not A and not B ”: $A^c \cap B^c = \{1, 2, 3\} \cap \{1, 3, 5\} = \{1, 3\}$. This is **De Morgan’s law**: $(A \cup B)^c = A^c \cap B^c$. Its twin, $(A \cap B)^c = A^c \cup B^c$, says “not both” means “at least one of them fails”. In Figure 2, the right panel is exactly the region outside both circles.

Extra: A Venn diagram shows overlap, not independence. Mutually exclusive events (see the [mutually exclusive events Note](#)) are circles that do not touch. Whether two overlapping events are independent needs a calculation: here $P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, but $P(A \cap B) = \frac{2}{6} = \frac{1}{3}$. The two differ, so A and B are dependent (see the [independent events Note](#)): knowing the roll is even makes “at least 4” more likely, $2/3$ instead of $1/2$.

3. Contingency tables

Key point: A contingency table puts the categories of one event in the rows and of the other in the columns; each inner cell counts the outcomes in one Venn region, and the totals count whole circles.

A contingency table (also called a crosstab, see the [bivariate and multivariate analysis Note](#)) counts every pair of categories of two columns. Applied to two events, it holds the same information as the Venn diagram, arranged as a grid.

3.1 The die as a table

Key point: Two yes/no events give a 2 by 2 table; its four inner cells are the four Venn regions.

For the die, the rows ask “is the number at least 4?” and the columns ask “is it even?”. Each cell counts the outcomes that answer both questions that way:

	even	odd	total
at least 4	2 (4, 6)	1 (5)	3
below 4	1 (2)	2 (1, 3)	3
total	3	3	6

Each inner cell matches one region of Figure 1: “at least 4 and even” is the overlap, “below 4 and odd” is the outside. The row totals count circle A and its complement; the column totals count circle B and its complement.

Each yes/no question works like a random variable with two values, 1 for yes and 0 for no (see the [random variables as functions Note](#)). The table lists every combination of the two.

3.2 A class of 100 students

Key point: When counts come from data, the table is filled from the facts given; the cell nobody mentions is what is left of the total.

A class has 100 students. 40 chose only Bio as their major, 30 chose only Maths, and 10 chose both. How many chose neither?

The table makes the missing count easy to find. The “neither” cell is whatever remains of the 100:

$$100 - 30 - 40 - 10 = 20$$

Figure 3 shows the counts both ways.

100 students: the same counts, two views

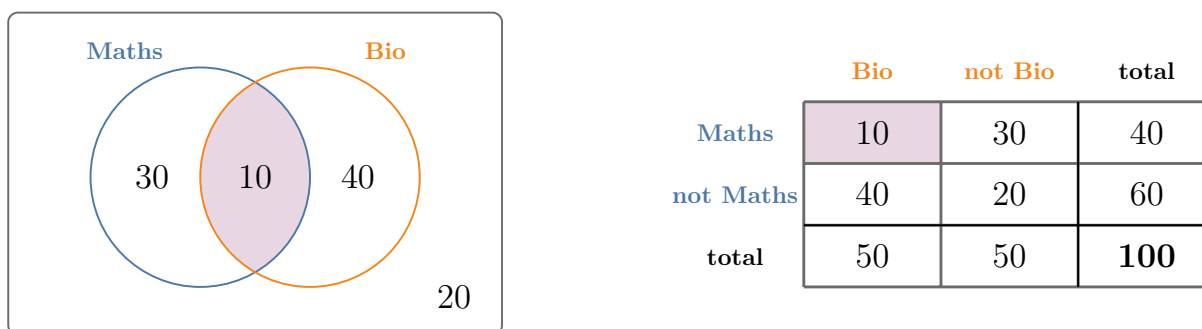


Figure 3: The 100 students as a Venn diagram (left) and a contingency table with totals (right)

Reading the table:

- **Inner cells:** 10 took both, 30 Maths only, 40 Bio only, 20 neither.
- **Row totals:** 40 took Maths (10 + 30), 60 did not.
- **Column totals:** 50 took Bio (10 + 40), 50 did not.
- **Grand total:** 100, the whole class.

The row and column totals are written in the margins of the table. They return in the joint, marginal and conditional probability. Note as marginal counts.

3.3 From counts to probabilities

Key point: Dividing every cell by the grand total turns the table into probabilities for one randomly chosen student.

Pick one student at random. Dividing each count by 100 gives the probability of each cell and each total:

	Bio	not Bio	total
Maths	0.10	0.30	0.40
not Maths	0.40	0.20	0.60
total	0.50	0.50	1.00

Now every probability from section 2 can be read from the table:

- $P(\text{Maths} \cap \text{Bio}) = 0.10$, one inner cell.
- $P(\text{Maths}) = 0.40$, a row total.
- $P(\text{Maths} \cup \text{Bio}) = 0.40 + 0.50 - 0.10 = 0.80$, the general addition rule.
- $P(\text{neither}) = 0.20 = 1 - 0.80$, the complement of the union.

4. Switching between the two views

Key point: Venn regions and table cells are the same sets of outcomes, so either view can be rebuilt from the other.

The Venn diagram and the contingency table carry exactly the same information:

Venn diagram	Contingency table
rectangle U	grand total
overlap of the circles	cell (yes, yes)
part of A outside B	cell (A yes, B no)
part of B outside A	cell (A no, B yes)
outside both circles	cell (no, no)
whole circle A	row total for A yes
whole circle B	column total for B yes

Each view suits different work:

- A **Venn diagram** shows how events overlap at a glance, and makes unions, complements and De Morgan's law easy to see.
- A **contingency table** scales to real data with many rows and more than two categories, such as three passenger classes. pandas builds it in one call, which the [joint, marginal and conditional probability Note](#) uses on the Titanic data.

5. Summary

Event	Venn region	Die (A : at least 4, B : even)	Students (Maths, Bio)
$A \cap B$	overlap	$\{4, 6\}$, $2/6$	0.10
$A \cup B$	inside either circle	$\{2, 4, 5, 6\}$, $4/6$	0.80
A^c	outside circle A	$\{1, 2, 3\}$, $3/6$	0.60
$(A \cup B)^c$	outside both	$\{1, 3\}$, $2/6$	0.20
U	whole rectangle	$\{1, \dots, 6\}$, 1	1

- The rectangle is the sample space; its probability is 1.
- Two events split the sample space into four regions, whose probabilities add to 1.
- A contingency table holds the same four regions as cells, and the circles as row and column totals.
- Dividing a table of counts by its grand total turns it into a table of probabilities.

6. Key terms

Term	Meaning
Venn diagram	A picture of events as circles inside a rectangle; overlaps show shared outcomes
Universal set (U)	The rectangle of a Venn diagram; in probability, the sample space, with $P(U) = 1$
De Morgan's law	$(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
Row and column totals	The sums in the margins of a contingency table; each counts one whole event
Grand total	The sum of every cell of a contingency table: the size of the whole sample